Q7) Laboratory Exercise:-

a) Answer the Following

Explain Selenium suite?

The **Selenium Suite** is a collection of tools used for automating web browsers, primarily for testing web applications. It provides a flexible and powerful environment for creating, executing, and managing automated test cases across different browsers and platforms.

The suite consists of four main components. **Selenium IDE (Integrated Development Environment)** is a browser extension that allows users to record and playback test cases. It is easy to use and suitable for beginners, as it does not require programming knowledge.

Selenium RC (Remote Control) was an earlier tool that enabled writing test scripts in multiple programming languages and controlling browsers through a server. However, it is now deprecated and has been replaced by WebDriver due to performance and architectural limitations.

Selenium WebDriver is the core component of the suite. It directly interacts with web browsers using browser-specific drivers and supports multiple programming languages such as Java, Python, and C#. It provides better speed, control, and flexibility, making it suitable for advanced and large-scale automation testing.

Selenium Grid is used for distributed and parallel test execution. It allows running test cases on multiple machines, browsers, and operating systems simultaneously, which is particularly useful for cross-browser testing and reducing overall test execution time.

What are the limitations of Selenium IDE?

Selenium IDE is useful for basic automation and learning purposes, but it has several limitations that restrict its use in advanced testing scenarios. It has **limited programming capability**, as it does not support complex logic such as loops, conditional statements, or dynamic decision-making. This makes it unsuitable for handling complex test cases.

It is also **browser-dependent**, since it works as a browser extension and is limited to supported browsers. Additionally, Selenium IDE is **not suitable for complex testing**, as it struggles with dynamic web elements, advanced validations, and integration with external systems such as databases.

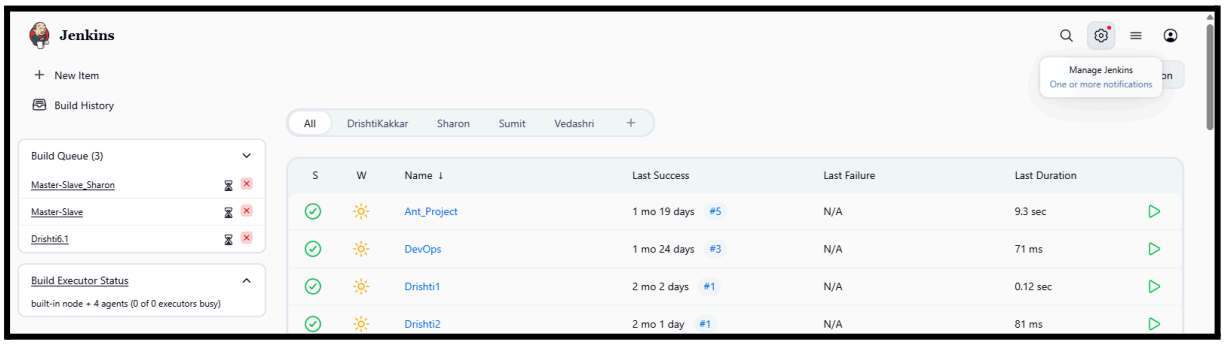
Another limitation is the **lack of robust error handling and debugging features**, which makes it difficult to identify and fix issues in test scripts. It also does not provide **native support for data-driven testing**, making it challenging to execute tests with multiple input datasets.

Furthermore, Selenium IDE has **limited integration capabilities** with testing frameworks like JUnit or TestNG, which are commonly used for structured automation testing. The recorded scripts are often fragile, leading to **maintenance issues** when there are changes

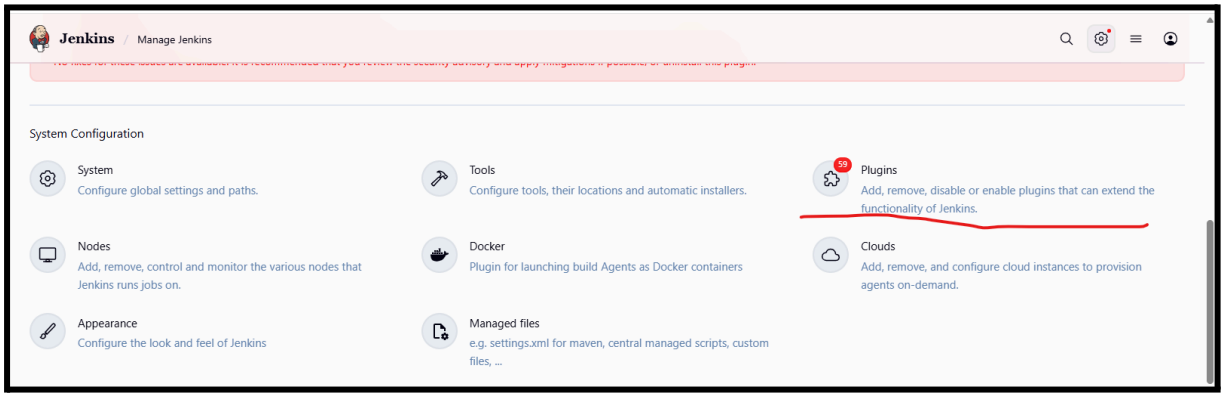
in the user interface. Finally, it does not support **parallel execution**, which increases the time required for running multiple test cases.

b) Execution

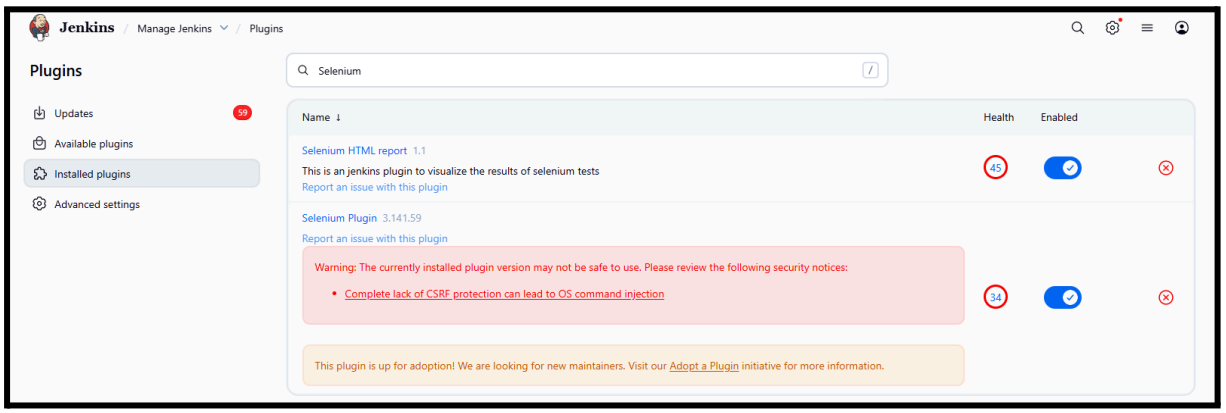
Jenkins Manage Jenkins



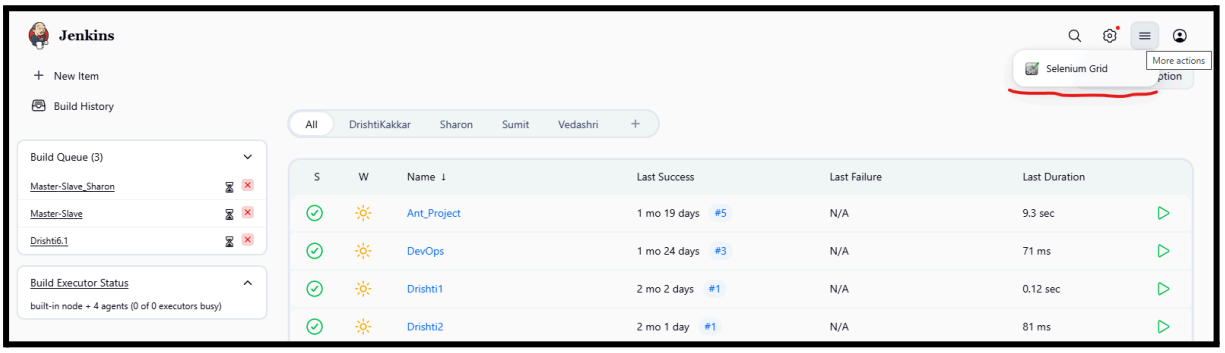
Click on plugins



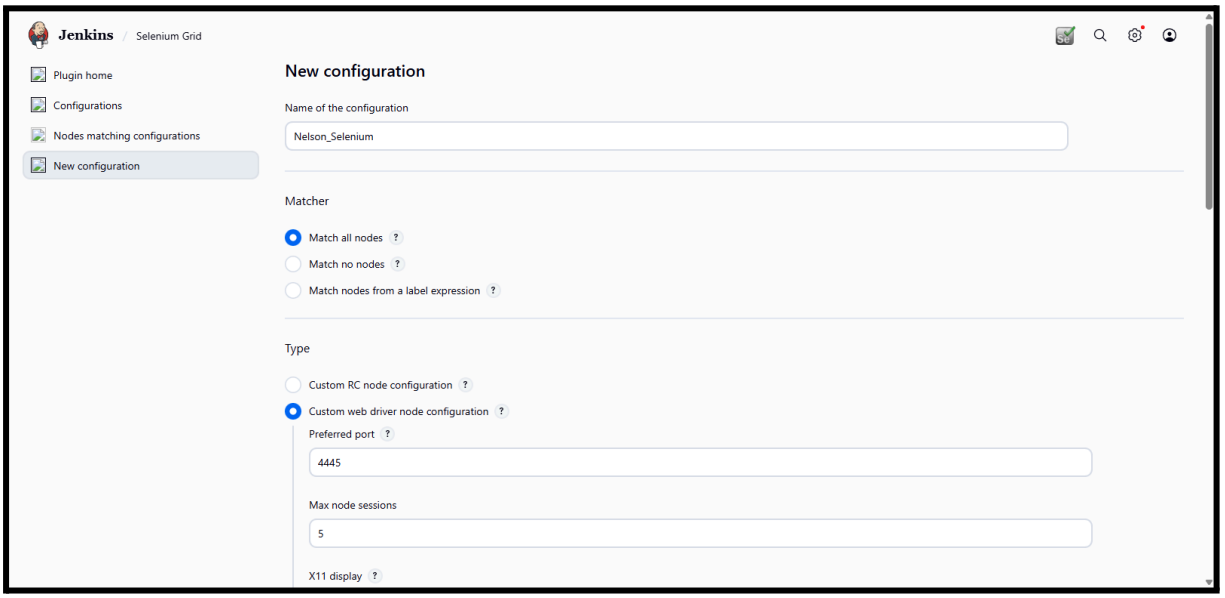
Install plugin : Selenium and Selenium HTML Report



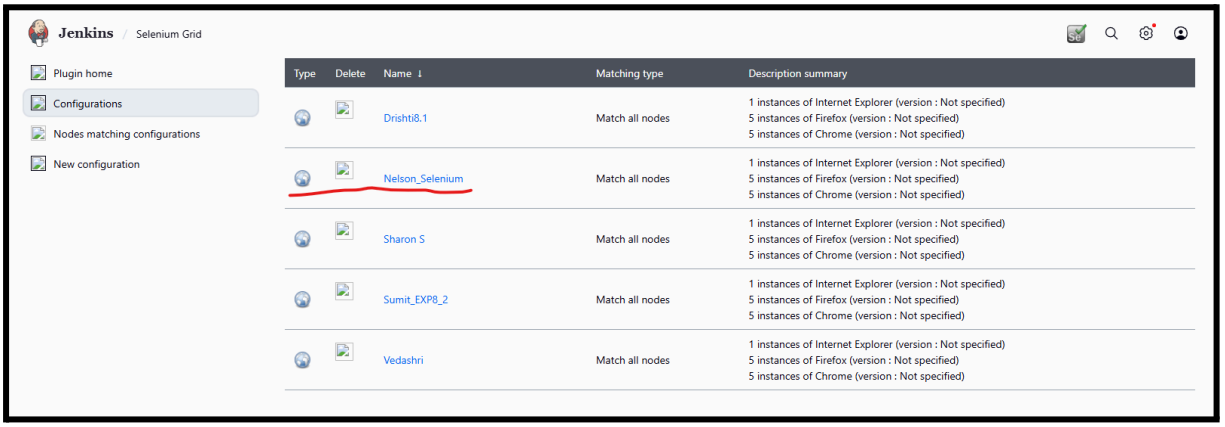
Jenkins More actions selenium grid



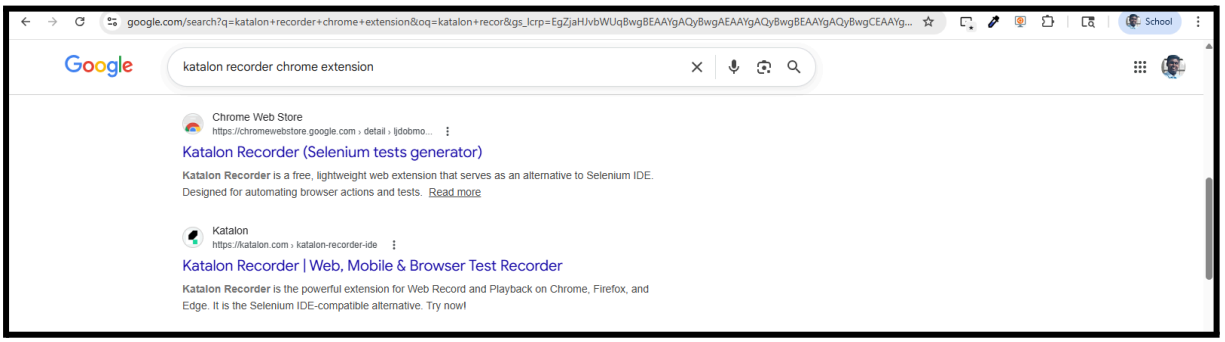
Click on selenium grid, configurations, new configuration, keep default settings , give name Vaishali J to configuration.



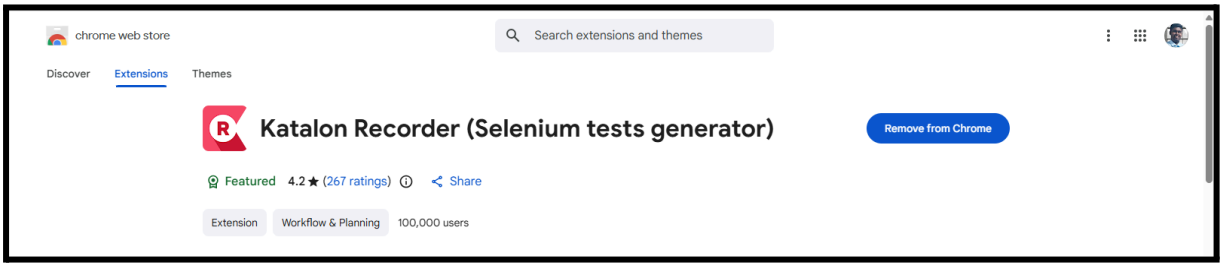
Save configuration

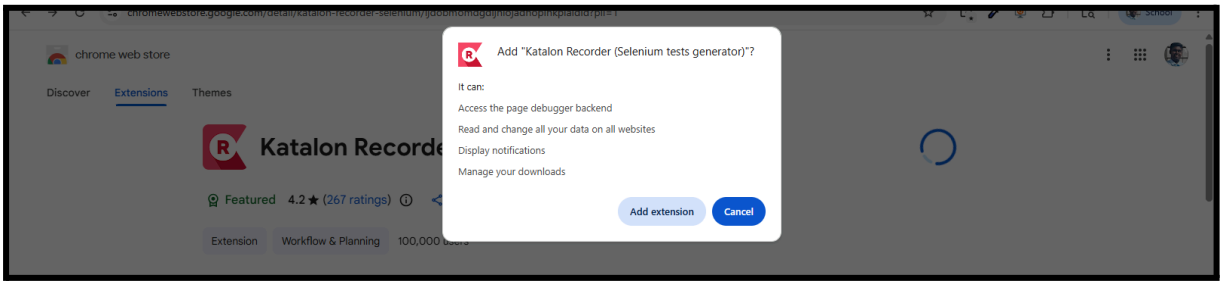


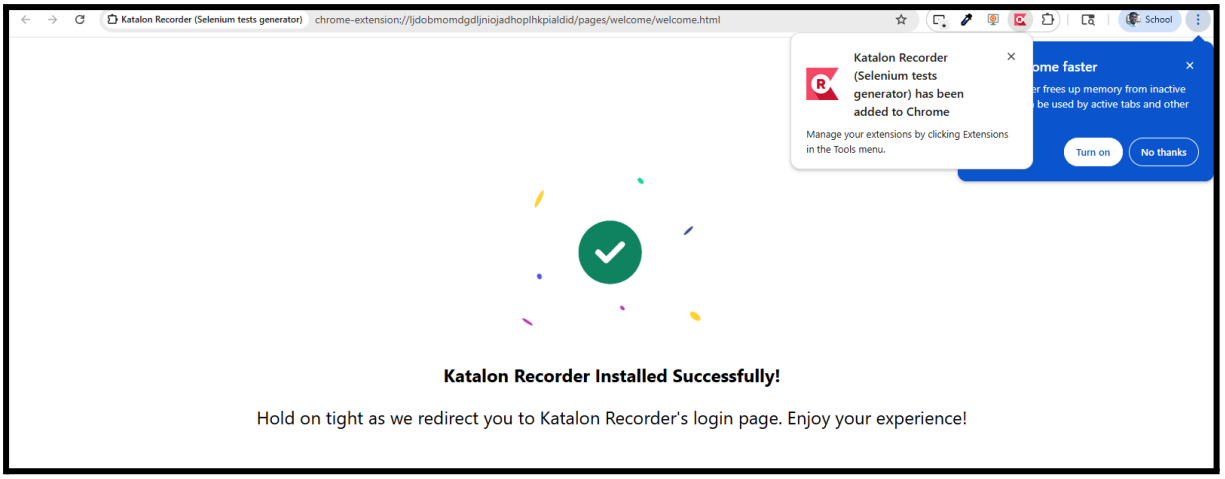
Google search Katalon Recorder



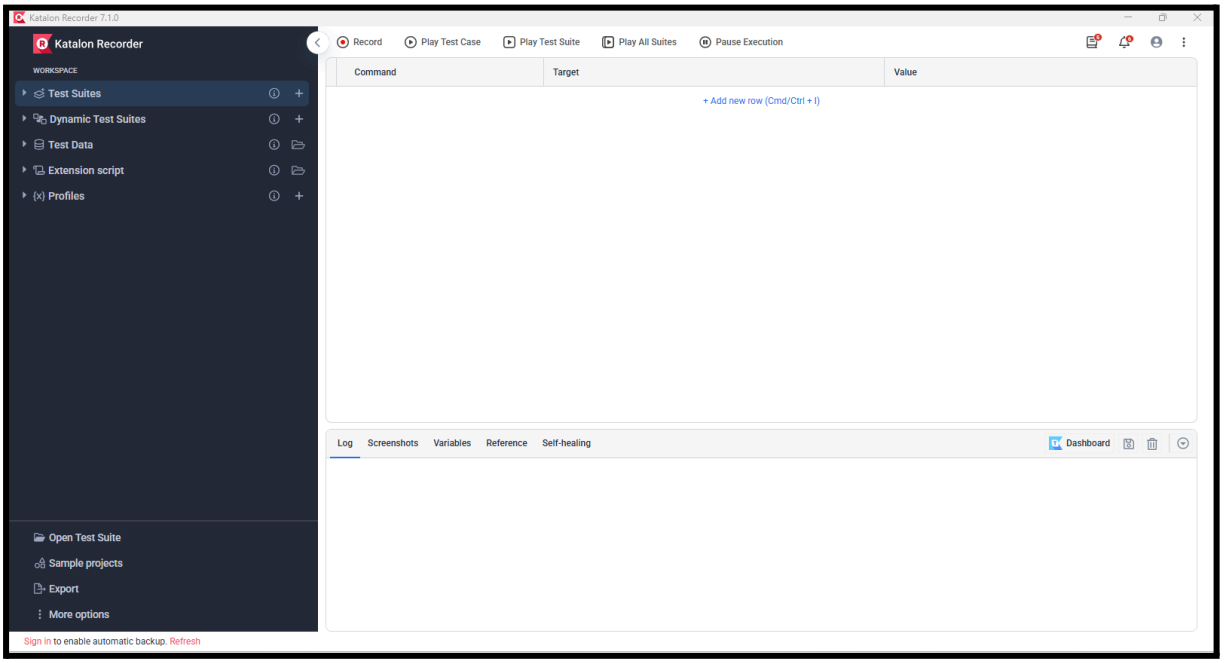
Add Extension, click on add to chrome



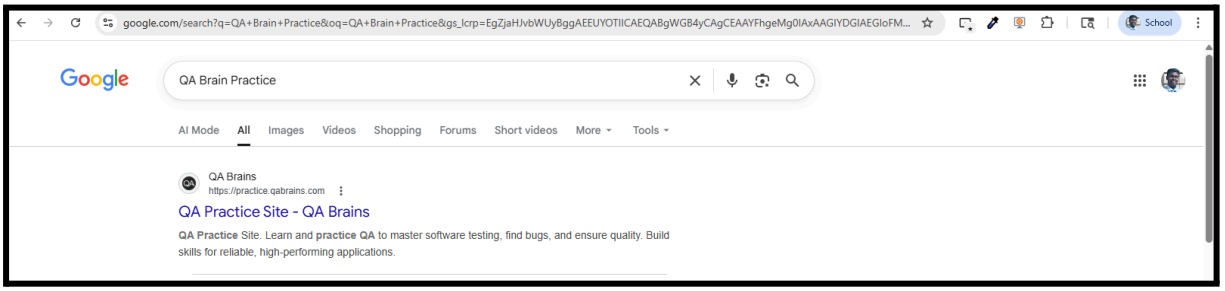




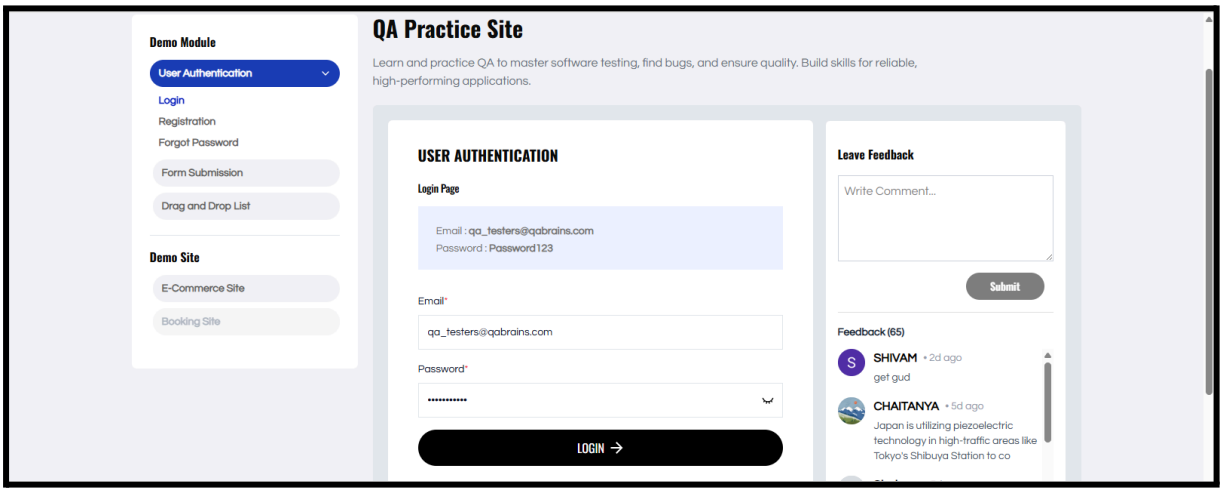
Katalon Recorder



Go to website QA Brain Practice site

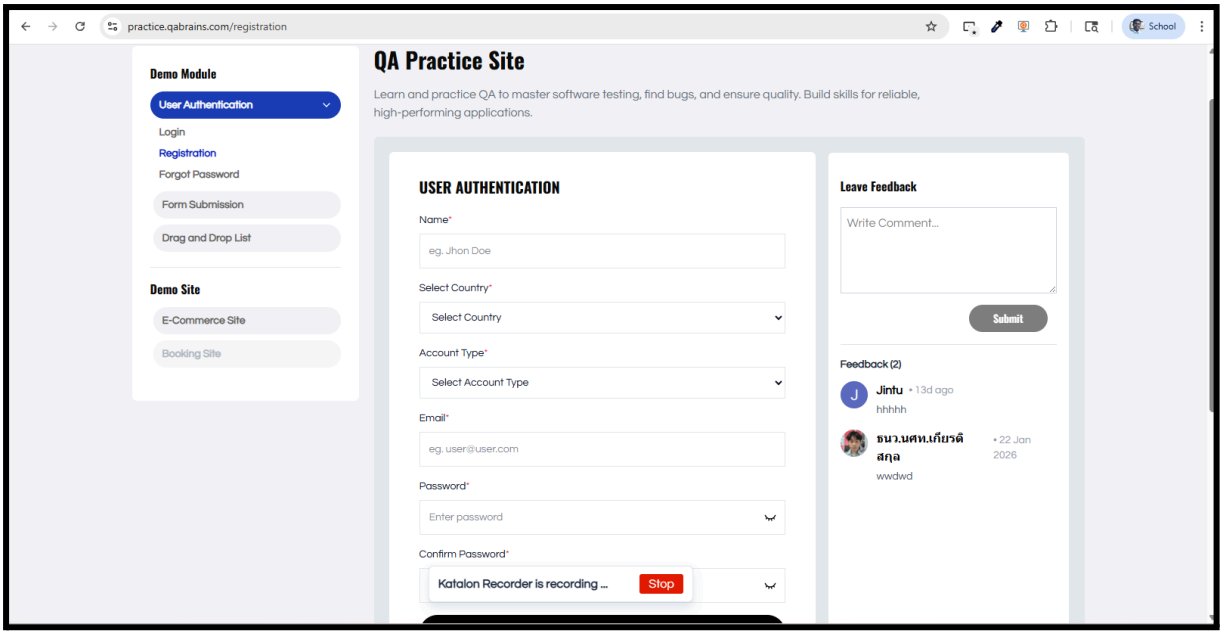


Login with given username and password

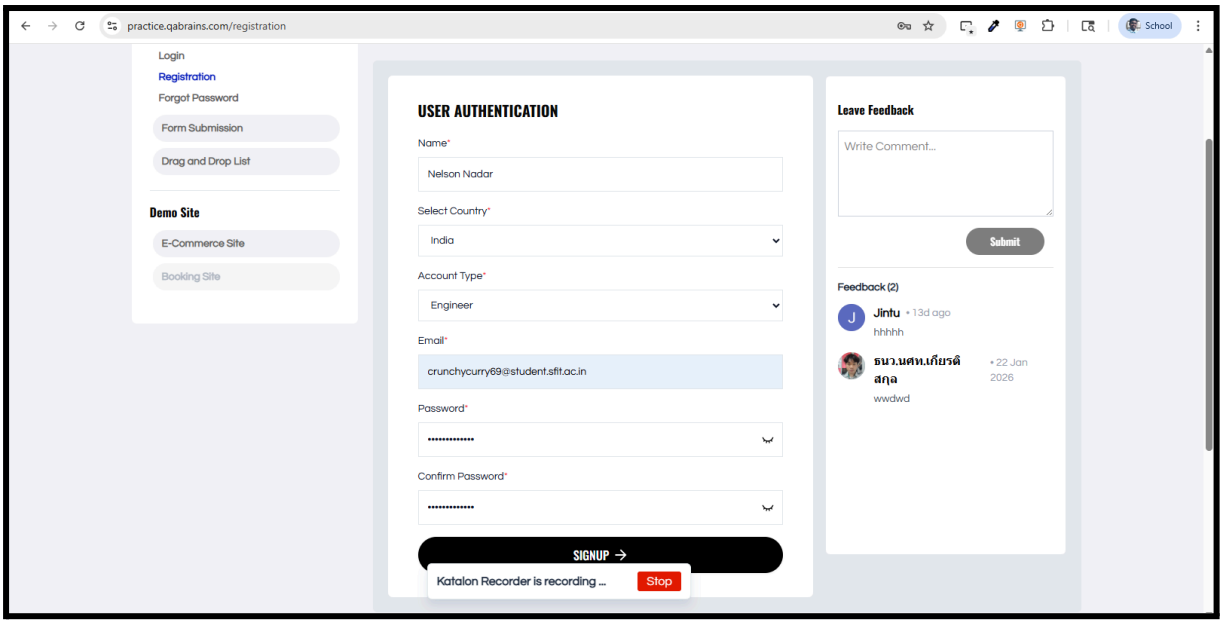


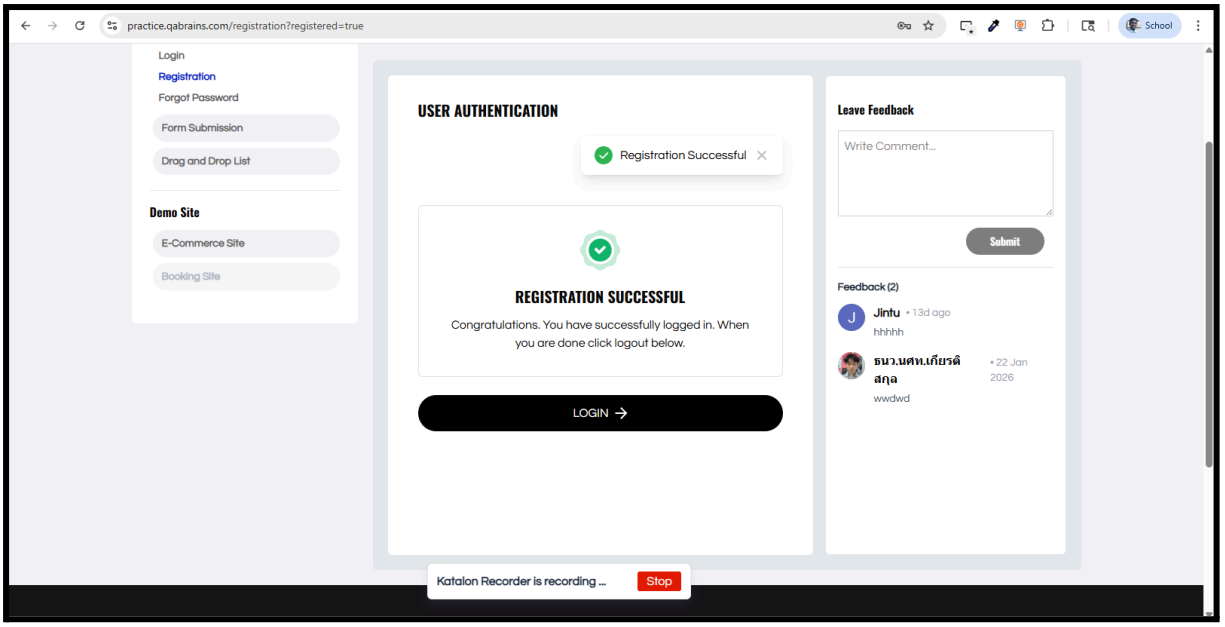
Go to Katalon and press record button

Click on Registration and fill all details



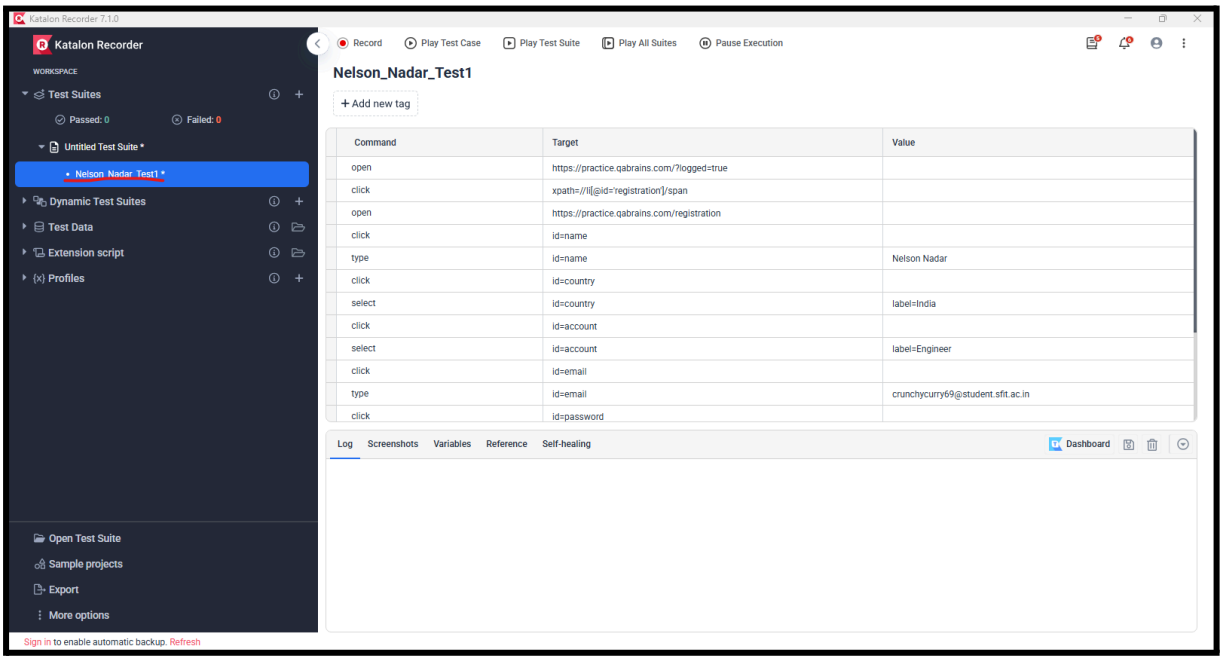
Make sure recording is on





Stop the recording and play test cases

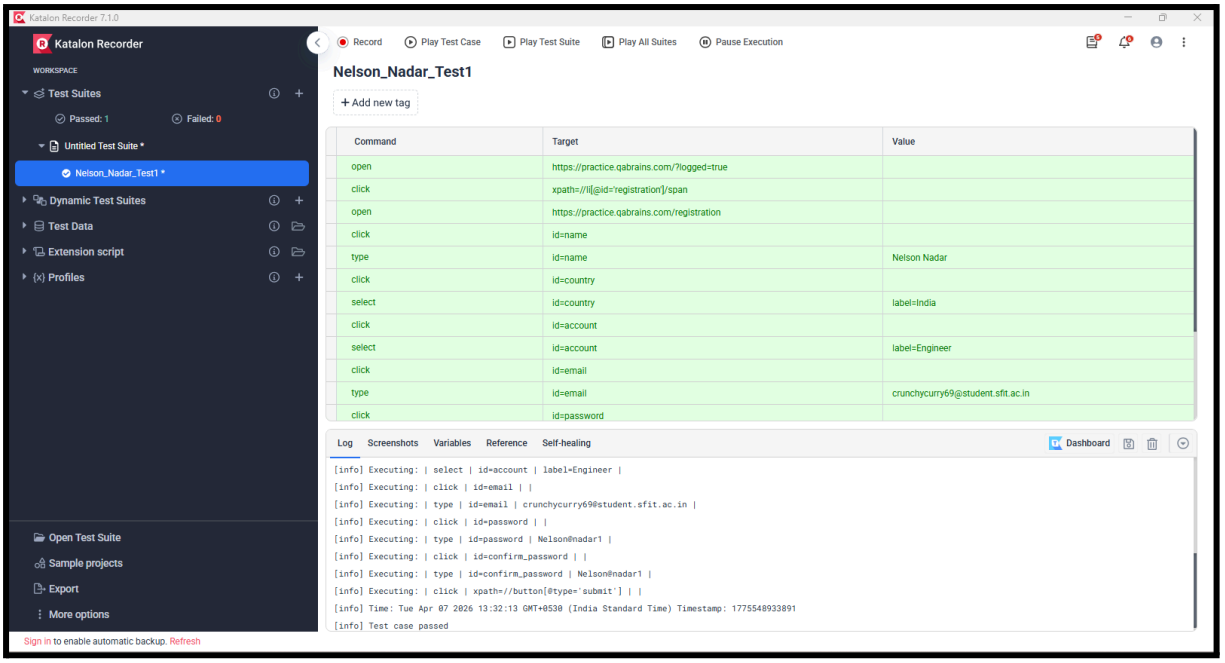
Check Katalon Recorder



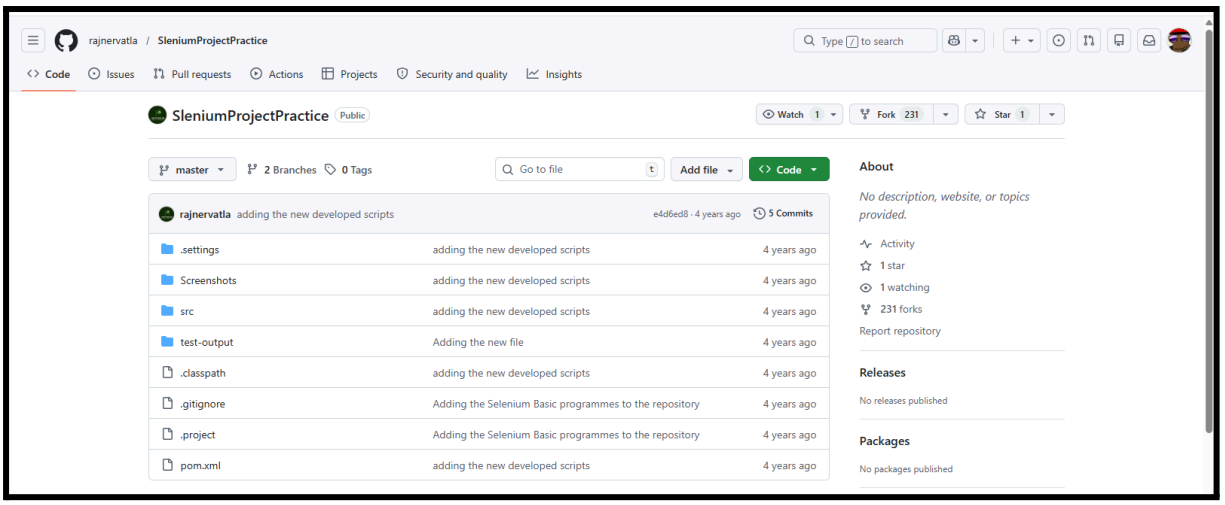
Click on play test cases

Check the output

Test case Passed

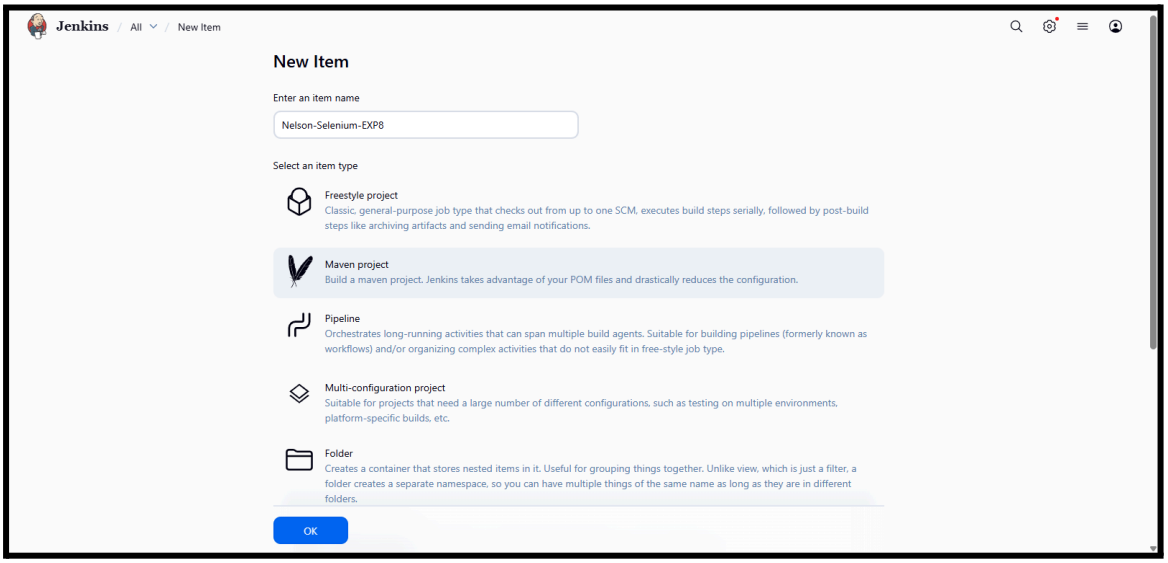


Go to Github and fork Selenium Project Practice repository

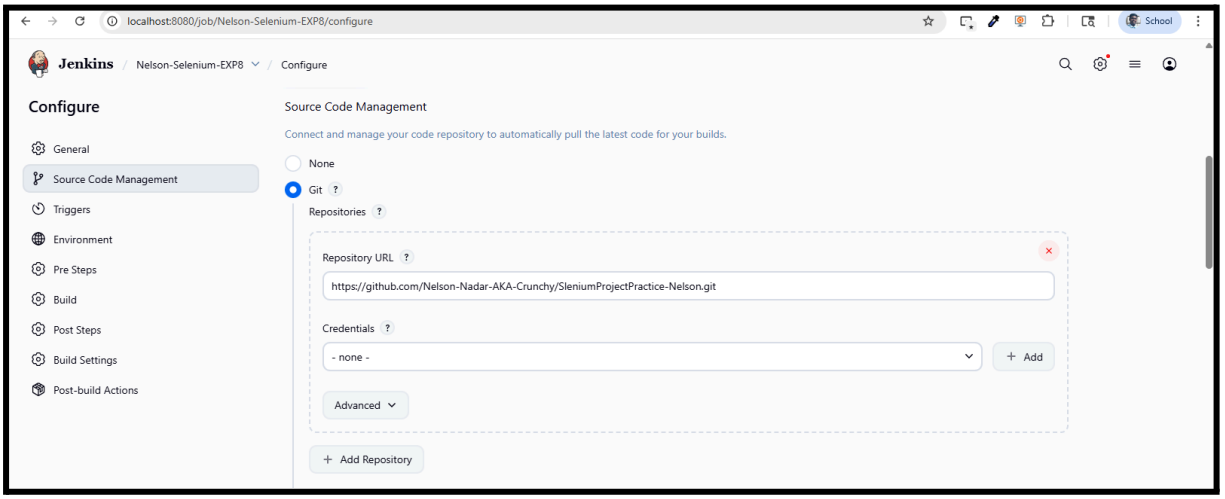


Copy URL

Go to Jenkins, New Project, Maven project



SCM : Git, Give Repository URL

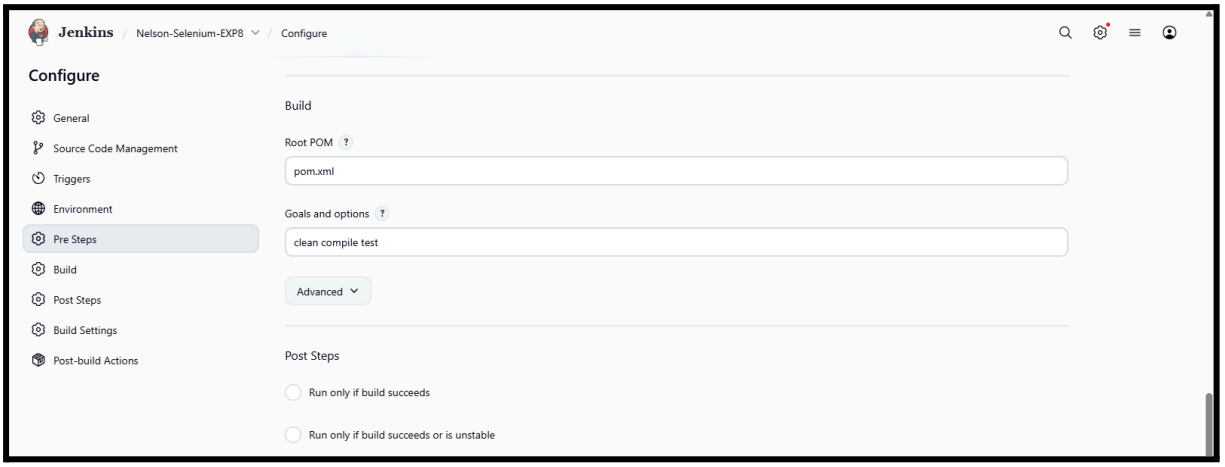


Build

Root POM : POM.XML

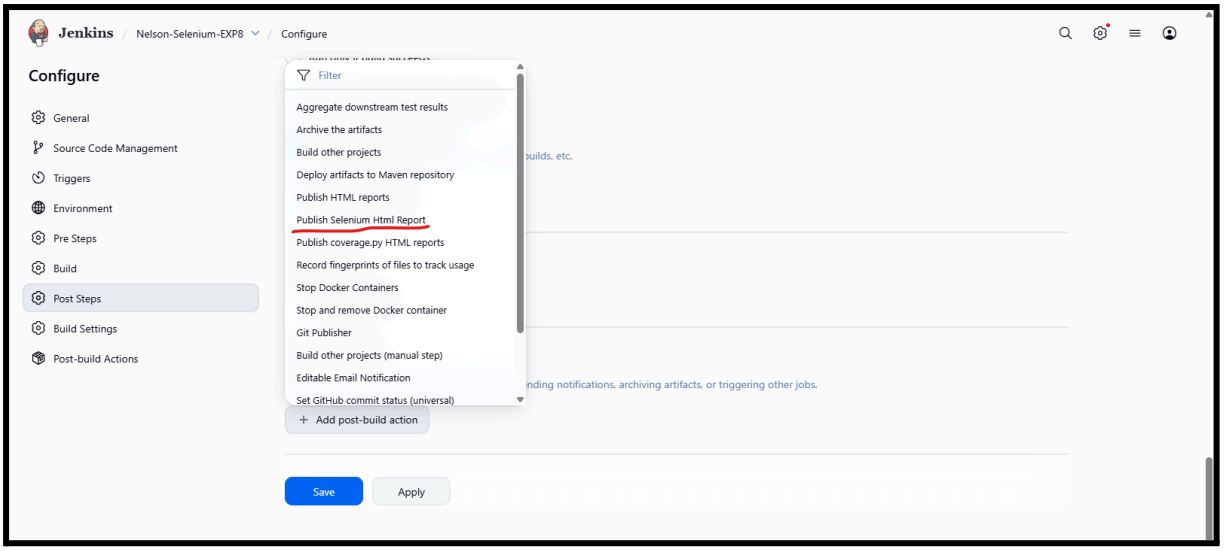
Goal and Options:

Clean, Compile and Test

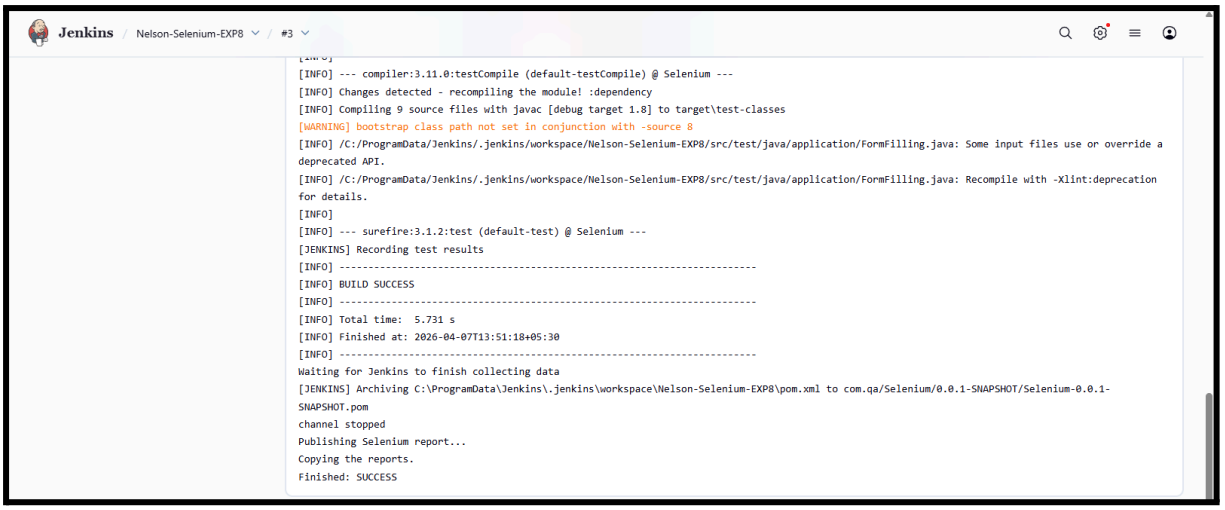
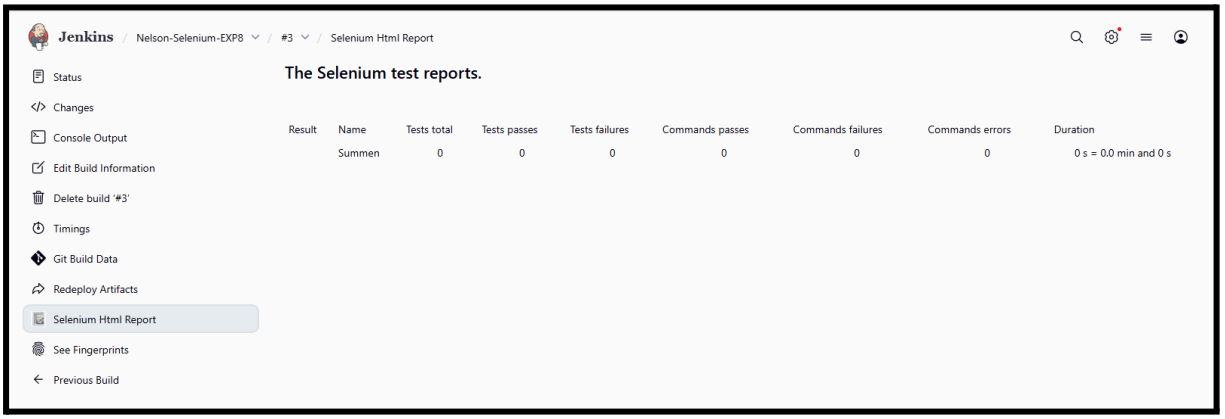


Post Build Action : Publish Selenium HTML Report

Apply and Save



Build now

Check Console output

And selenium HTML Report